
Adaptive Conformal Prediction for Chaotic Time Series

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Abstract

1 Introduction

Chaotic dynamical systems, characterized by extreme sensitivity to initial conditions, pose fundamental challenges for long-horizon forecasting. While neural networks such as LSTMs achieve impressive short-term accuracy, deterministic chaos causes small errors to amplify exponentially [Lorenz, 1963, Strogatz, 2015]. This challenge intensifies in *closed-loop forecasting*, where multi-step trajectories must be generated once and committed, without retroactive modification as new observations arrive. Unlike open-loop forecasting, which incorporates true observations at each step, closed-loop prediction allows errors to compound unchecked. Point forecast metrics such as RMSE quantify average error but fail to capture the growing uncertainty inherent in chaotic predictions, motivating the need for rigorous *uncertainty quantification* (UQ) with formal coverage guarantees.

Conformal prediction [Vovk et al., 2005, Shafer and Vovk, 2008] provides distribution-free UQ, guaranteeing marginal coverage under exchangeable data. However, closed-loop forecasting on chaotic systems fundamentally violates this assumption: as the model feeds predictions back as input, the induced distribution drifts from the calibration distribution due to accumulated errors and chaotic sensitivity. Empirical coverage degrades rapidly with forecast horizon, often dropping 20–30% below the nominal $(1 - \alpha)$ level within moderate horizons.

We conduct a systematic empirical study of conformal prediction under closed-loop chaotic forecasting using the Lorenz 63 system [Lorenz, 1963] as a testbed. We quantify coverage degradation as a function of horizon and evaluate *adaptive conformal calibration* strategies—rolling-window and exponentially weighted moving average (EWMA) schemes—that recalibrate *uncertainty estimates* to track distribution shift. Our analysis reveals when and where conformal intervals fail and provides practical guidance for nonstationary time series.

Contributions We demonstrate that split conformal prediction suffers severe coverage degradation (20–30% below nominal) under closed-loop chaotic forecasting, and show that adaptive methods (rolling window, EWMA) substantially recover coverage at the cost of wider intervals. We provide empirical insights into the coverage–width trade-off and identify spatial heterogeneity in failure modes across the Lorenz attractor.

2 Problem Setup

2.1 Chaotic Time Series Forecasting Task

We consider multi-step-ahead forecasting in chaotic dynamical systems. Let $x_t \in \mathbb{R}^d$ denote the system state at time t . Given a history window $\{x_{t-k+1}, \dots, x_t\}$ of length k , the task is to predict

future states $\{x_{t+1}, \dots, x_{t+h}\}$ over horizon h . Chaotic systems exhibit sensitive dependence on initial conditions: small perturbations grow exponentially, causing prediction errors to accumulate rapidly with horizon h . This motivates rigorous uncertainty quantification for reliable long-horizon forecasting.

2.2 The Lorenz 63 System as a Testbed

We use the Lorenz 63 system [Lorenz, 1963] as our testbed. The equations are:

$$\frac{dx}{dt} = \sigma(y - x), \quad \frac{dy}{dt} = x(\rho - z) - y, \quad \frac{dz}{dt} = xy - \beta z, \quad (1)$$

where $(x, y, z) \in \mathbb{R}^3$ is the state. With standard parameters $\sigma = 10$, $\rho = 28$, and $\beta = 8/3$, the system exhibits deterministic chaos on a butterfly-shaped attractor. The Lorenz system is ideal for our study: it is low-dimensional (enabling visualization), exhibits clear chaotic behavior, and has well-characterized dynamics.¹

2.3 Open-loop vs Closed-loop Prediction

We distinguish two forecasting protocols that differ fundamentally in how they handle sequential predictions. In *closed-loop forecasting*, model predictions are recursively fed back as inputs during rollout, while ground-truth observations are not used to update the forecasted trajectory. This contrasts with *open-loop forecasting*, where the model predicts each future state x_{t+h} using the true observation at the previous step:

$$\hat{x}_{t+h}^{\text{open}} = \hat{f}(x_{t+h-1}), \quad (2)$$

where x_{t+h-1} is the true observed state. In closed-loop forecasting, the model recursively feeds its own predictions back as inputs:

$$\hat{x}_{t+1}^{\text{closed}} = \hat{f}(x_t), \quad \hat{x}_{t+2}^{\text{closed}} = \hat{f}(\hat{x}_{t+1}^{\text{closed}}), \quad \dots, \quad \hat{x}_{t+h}^{\text{closed}} = \hat{f}(\hat{x}_{t+h-1}^{\text{closed}}). \quad (3)$$

This distinction is crucial for uncertainty quantification. In open-loop forecasting, each prediction operates on the true data distribution, and errors do not propagate between steps. In closed-loop forecasting, model errors accumulate and the effective input distribution drifts from the training distribution as horizon increases. Note that model training uses standard one-step-ahead supervised learning with teacher forcing, and is independent of the closed-loop forecasting setup. The closed-loop feedback loop fundamentally alters the error characteristics and poses unique challenges for calibrating prediction intervals, which we address in subsequent sections.

3 Conformal Prediction Methods

3.1 Motivation for Adaptive Conformal Inference

Standard conformal prediction relies on the exchangeability of calibration and test data. Under this assumption, the empirical quantile computed on the calibration set yields finite-sample marginal coverage guarantees [Vovk et al., 2005, Shafer and Vovk, 2008].

However, in sequential forecasting (particularly in closed-loop rollouts), prediction errors can evolve over time due to error accumulation and model misspecification. As the model recursively feeds its own predictions back as inputs, the effective input distribution drifts progressively from the calibration distribution, violating exchangeability and leading to coverage degradation.

To address this issue, Gibbs and Candès [2021] proposed adaptive conformal inference (ACI), which dynamically updates the conformal quantile at each time step to maintain coverage under distribution shift. Rather than relying on a fixed calibration set, ACI continuously re-estimates the quantile by treating the distribution shift as an online learning problem in a single adaptive parameter. In our work, we apply this principle to the closed-loop chaotic forecasting setting, an application domain not previously studied in the adaptive conformal literature.

¹Additional system properties are detailed in Appendix A.1.

3.2 Split Conformal Prediction for Time Series

Split conformal prediction [Lei et al., 2018, Romano et al., 2019] constructs prediction intervals with finite-sample coverage guarantees under exchangeability. Given a trained forecasting model $\hat{f}$ and a target coverage level $1 - \alpha$, the method operates as follows.

First, we partition the available data into a training set (used to train $\hat{f}$), a calibration set $\mathcal{D}_{\text{cal}} = \{(x_i, y_i)\}_{i=1}^n$, and a test set. For time series, the calibration set consists of a contiguous historical window.

Second, we compute *nonconformity scores* on the calibration set, measuring the discrepancy between predictions and observations:

$$R_i = \|y_i - \hat{f}(x_i)\|, \quad i = 1, \dots, n, \quad (4)$$

where we use the Euclidean norm for the multivariate Lorenz system.

Third, we compute the $(1 - \alpha)$ -quantile of the calibration scores:

$$\hat{q}_{1-\alpha} = \text{Quantile}_{1-\alpha}(\{R_1, \dots, R_n\}), \quad (5)$$

defined as the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest score to ensure marginal coverage [Vovk et al., 2005].

Finally, for a new test point x_{test} , the prediction interval is:

$$C(x_{\text{test}}) = [\hat{f}(x_{\text{test}}) - \hat{q}_{1-\alpha}, \hat{f}(x_{\text{test}}) + \hat{q}_{1-\alpha}]. \quad (6)$$

Under exchangeability of $(x_{\text{test}}, y_{\text{test}})$ with $\mathcal{D}_{\text{cal}}$, this interval achieves $\mathbb{P}(y_{\text{test}} \in C(x_{\text{test}})) \geq 1 - \alpha$.

3.3 Limitations under Distribution Shift

The coverage guarantee critically relies on *exchangeability*: calibration and test data must be drawn from the same distribution. Closed-loop forecasting systematically violates this assumption.

Distribution Drift in Closed-Loop Forecasting. In open-loop forecasting, the model predicts $\hat{x}_{t+1}$ using the true observation x_t as input. Each prediction step is independent, and residuals remain approximately stationary. In closed-loop forecasting, the model recursively consumes its own predictions:

$$\hat{x}_{t+1} = \hat{f}(x_t), \quad \hat{x}_{t+2} = \hat{f}(\hat{x}_{t+1}), \quad \hat{x}_{t+3} = \hat{f}(\hat{x}_{t+2}), \quad \dots \quad (7)$$

Even if the initial prediction $\hat{x}_{t+1}$ has a small error ϵ_1 , this error propagates and amplifies over subsequent steps due to chaotic sensitivity. The effective input distribution at time $t + h$ drifts progressively from the calibration distribution as h increases.

Breakdown of Exchangeability. As the forecast horizon grows, the residual distribution $p(y_{t+h} - \hat{y}_{t+h})$ becomes increasingly heavy-tailed and location-shifted relative to the calibration residuals. The calibration quantile $\hat{q}_{1-\alpha}$ —computed from open-loop or early-horizon residuals—underestimates the true uncertainty at later horizons. Consequently, empirical coverage degrades:

$$\mathbb{P}(y_{t+h} \in C(x_{t+h})) \ll 1 - \alpha \quad \text{for large } h. \quad (8)$$

This breakdown reflects violation of the foundational exchangeability assumption, not a failure of the conformal framework itself.

3.4 Adaptive Conformal Calibration

To mitigate distribution shift, we employ *adaptive conformal calibration*, which recalibrates the quantile $\hat{q}_{1-\alpha}$ as new observations arrive. *Ground-truth observations are used solely for uncertainty recalibration after predictions are made, and do not alter the closed-loop forecasted trajectory.* The key idea is to replace the fixed calibration set with a dynamically tracked window of recent residuals.

Algorithm 1 Adaptive Conformal Prediction (Rolling Window)

Require: Trained model $\hat{f}$, window size W , coverage level $1 - \alpha$, initial calibration set $\mathcal{D}_{\text{cal}}$
Require: Test sequence $\{x_t\}_{t=1}^T$
1: Initialize buffer $\mathcal{B} \leftarrow \{(x_i, y_i, R_i)\}_{i \in \mathcal{D}_{\text{cal}}}$ of size W
2: **for** $t = 1$ to T **do**
3: Compute prediction $\hat{y}_t = \hat{f}(x_t)$
4: Compute quantile $\hat{q}_{1-\alpha}^{(t)} = \text{Quantile}_{1-\alpha}(\{R_i : (x_i, y_i, R_i) \in \mathcal{B}\})$
5: Output interval $C_t = [\hat{y}_t - \hat{q}_{1-\alpha}^{(t)}, \hat{y}_t + \hat{q}_{1-\alpha}^{(t)}]$
6: Observe true value y_t $\triangleright$ Used only for recalibration, not trajectory update
7: Compute residual $R_t = \|y_t - \hat{y}_t\|$
8: Update buffer: $\mathcal{B} \leftarrow \mathcal{B} \setminus \{(x_{t-W}, y_{t-W}, R_{t-W})\} \cup \{(x_t, y_t, R_t)\}$
9: **end for**

Rolling Window Strategy. At each test time t , we maintain a rolling window of the most recent W observations and their residuals:

$$\mathcal{D}_{\text{rolling}}(t) = \{(x_{t-W}, y_{t-W}), \dots, (x_{t-1}, y_{t-1})\}. \quad (9)$$

We compute the time-dependent quantile:

$$\hat{q}_{1-\alpha}^{(t)} = \text{Quantile}_{1-\alpha}(\{R_{t-W}, \dots, R_{t-1}\}), \quad (10)$$

and construct the prediction interval $C_t(x_t) = [\hat{y}_t - \hat{q}_{1-\alpha}^{(t)}, \hat{y}_t + \hat{q}_{1-\alpha}^{(t)}]$. This approach is reactive: if the error distribution shifts (e.g., due to entering a new region of the attractor or error accumulation), the quantile is recalibrated accordingly. Smaller W enables faster adaptation but increases variance in $\hat{q}_{1-\alpha}^{(t)}$ due to fewer calibration points; larger W provides stability but slower tracking.

Exponentially Weighted Moving Average (EWMA). Alternatively, we weight recent residuals exponentially using decay parameter $\lambda \in (0, 1)$:

$$\hat{q}_{1-\alpha}^{(t)} = \text{WeightedQuantile}_{1-\alpha}(\{R_{t-k}\}_{k=1}^{\infty}, \{w_k = \lambda^k\}_{k=1}^{\infty}), \quad (11)$$

where the weighted quantile is computed by sorting residuals and finding the smallest value such that cumulative normalized weight exceeds $1 - \alpha$. In practice, we truncate history to $k \leq 500$ and normalize weights to sum to 1. Values $\lambda \approx 1$ (e.g., 0.99) prioritize stability, while smaller λ (e.g., 0.90) prioritize reactivity.

Update Frequency. Both rolling and EWMA methods require recomputing the quantile at each time step, which can be computationally expensive for large W or long test sequences. In practice, we update the quantile every $k = 5$ steps rather than every step, using the same $\hat{q}_{1-\alpha}^{(t)}$ for multiple consecutive predictions. This reduces computation by 80% while maintaining reasonable tracking performance, as the error distribution typically does not shift drastically within 5 steps.

Algorithm 1 summarizes the rolling window procedure; EWMA follows an analogous update rule.²

Coverage–Width Trade-off. Adaptive methods face a fundamental trade-off. When the error distribution shifts upward (e.g., due to accumulated forecast error), adaptive conformal widens intervals to maintain coverage, increasing average width compared to split conformal. However, split conformal’s narrower intervals come at the cost of severe undercoverage. For safety-critical applications, we argue that reliable coverage with adaptive, wider intervals is preferable to narrow intervals with poor coverage guarantees.

4 Results

We present empirical results on the Lorenz 63 system with measurement noise $\sigma_{\text{noise}} = 0.5$. All experiments target coverage $1 - \alpha = 0.90$ and evaluate forecast horizons $h \in \{10, 50, 100\}$. Our

²Parameter selection heuristics are detailed in Appendix A.3.

results demonstrate severe coverage degradation of split conformal prediction under closed-loop forecasting and show that adaptive calibration substantially recovers reliable uncertainty quantification.

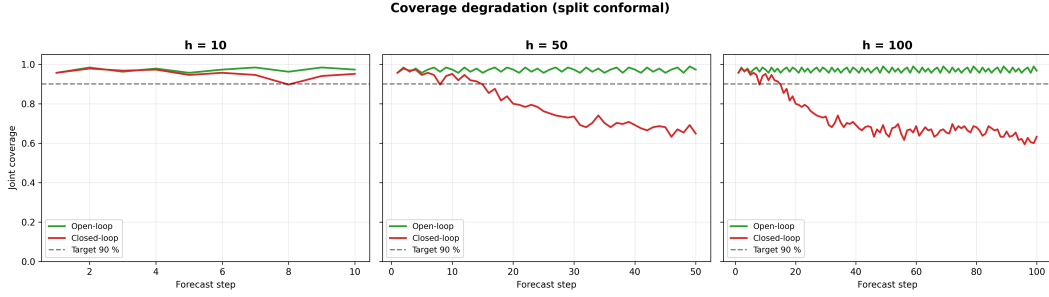


Figure 1: Coverage degradation in closed-loop forecasting with split conformal prediction. Each panel shows coverage evolution across forecast steps for different horizons ($h \in \{10, 50, 100\}$). Green lines indicate open-loop forecasting (near-nominal coverage), while red lines show closed-loop forecasting where coverage degrades progressively due to distribution shift.

4.1 Coverage Degradation in Closed-loop Forecasting

Figure 1 shows empirical coverage as a function of forecast step for split conformal prediction. Under open-loop evaluation, coverage remains near-nominal (≈ 0.97) across all horizons, validating the conformal calibration when model inputs follow the true data distribution. In contrast, closed-loop rollout leads to progressive coverage degradation as forecast horizon increases.

At short horizons ($h = 10$), closed-loop coverage remains close to nominal (0.951, only 2% below open-loop). However, degradation becomes substantial at longer horizons: coverage drops to 0.792 at $h = 50$ (18% degradation) and collapses to 0.723 at $h = 100$ (26% degradation), falling far below the target level.

4.2 Adaptive Conformal vs Standard Conformal

Adaptive conformal methods substantially mitigate coverage degradation by updating calibration quantiles to track the evolving error distribution. Figure 2 and Table 1 summarize performance across all methods.

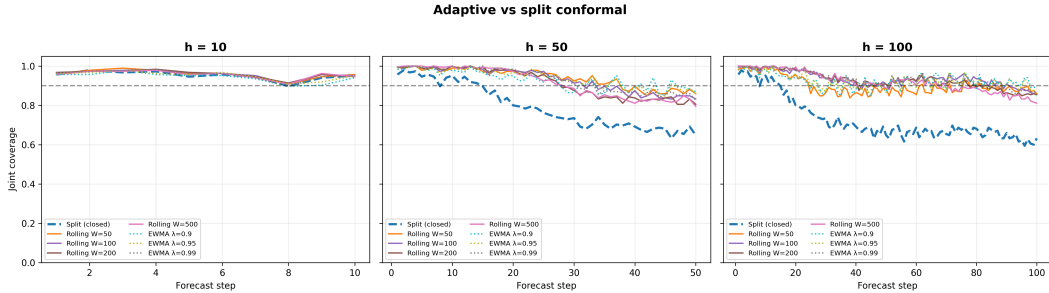


Figure 2: Comparison of adaptive conformal methods versus split conformal across horizons. The blue dashed line represents split conformal in closed-loop, showing severe degradation at longer horizons. Rolling window (solid lines) and EWMA methods (dotted lines) maintain coverage near the target 0.90 level by adaptively updating quantiles.

At $h = 10$, all methods achieve comparable coverage, as distribution shift remains modest. At longer horizons ($h = 50$ and $h = 100$), split conformal exhibits pronounced undercoverage, while rolling-window and EWMA methods maintain coverage near the nominal level by adapting to error

Table 1: Coverage and interval width for split and adaptive conformal methods in closed-loop forecasting. Joint coverage reports the fraction of test points where all three dimensions simultaneously fall within prediction intervals. Target coverage: $1 - \alpha = 0.90$.

Method	Joint Coverage			Width		
	$h = 10$	$h = 50$	$h = 100$	$h = 10$	$h = 50$	$h = 100$
Split Conformal	0.951	0.792	0.723	2.72	2.72	2.72
Rolling ($W = 50$)	0.959	0.941	0.900	2.82	4.85	6.47
Rolling ($W = 100$)	0.959	0.931	0.935	2.80	5.11	7.36
Rolling ($W = 200$)	0.959	0.915	0.929	2.78	5.41	7.78
Rolling ($W = 500$)	0.956	0.916	0.921	2.76	5.25	7.63
EWMA ($\lambda = 0.90$)	0.943	0.930	0.924	2.74	4.34	5.42
EWMA ($\lambda = 0.95$)	0.952	0.937	0.923	2.76	4.63	6.10
EWMA ($\lambda = 0.99$)	0.956	0.933	0.934	2.77	5.18	7.43

accumulation. Across configurations, adaptive methods recover most of the lost coverage at long horizons, with more reactive settings yielding narrower but more variable intervals.

4.3 Adaptive Conformal Calibration: Mechanisms and Trade-offs

Quantile Evolution under Closed-loop Forecasting. The improved coverage of adaptive methods is driven by their ability to dynamically track the evolving error distribution. Figure 3 illustrates the evolution of the nonconformity quantile $\hat{q}_{1-\alpha}$ throughout the forecast sequence.

At short horizons ($h = 10$), all methods maintain quantiles near the split conformal baseline ($\hat{q}_{1-\alpha} = 1.36$), reflecting minimal distribution shift. As the horizon increases to $h = 50$, adaptive quantiles begin to diverge from this baseline, expanding to accommodate growing uncertainty. Rolling window methods exhibit step-like adjustments as new high-error observations enter the buffer, whereas EWMA methods show smoother temporal evolution.

At $h = 100$, the divergence becomes substantial: while split conformal’s fixed quantile remains at 1.36, adaptive methods expand their quantiles to approximately 3.5–4.0 by the end of the forecast sequence—nearly three times the split conformal value. This expansion directly captures the increased variability induced by error accumulation and chaotic sensitivity. More reactive configurations ($W = 50$, $\lambda = 0.90$) respond rapidly but exhibit greater variability, whereas more conservative settings ($W = 500$, $\lambda = 0.99$) adjust more smoothly at the cost of delayed response.

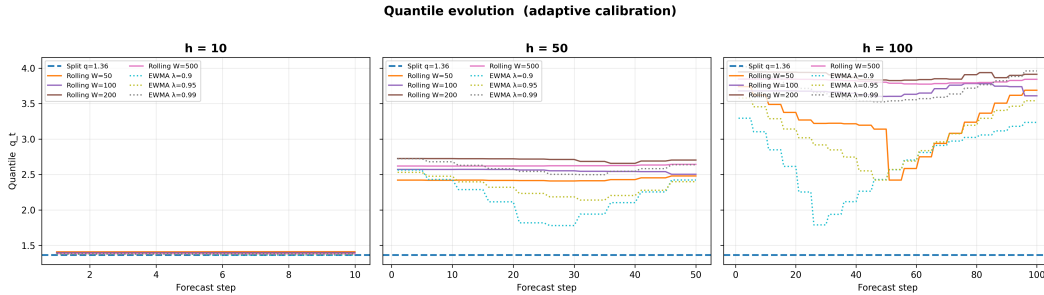


Figure 3: Evolution of the conformity quantile $\hat{q}_{1-\alpha}$ across forecast steps for different adaptive methods. The blue dashed line shows the fixed split conformal quantile ($q = 1.36$), which remains constant throughout all horizons. Rolling window methods (solid lines) and EWMA methods (dotted lines) dynamically adjust the quantile in response to the changing error distribution. At $h = 100$, the quantile expands significantly (reaching 3.5–4.0) to accommodate the increased uncertainty from error accumulation and distribution shift.

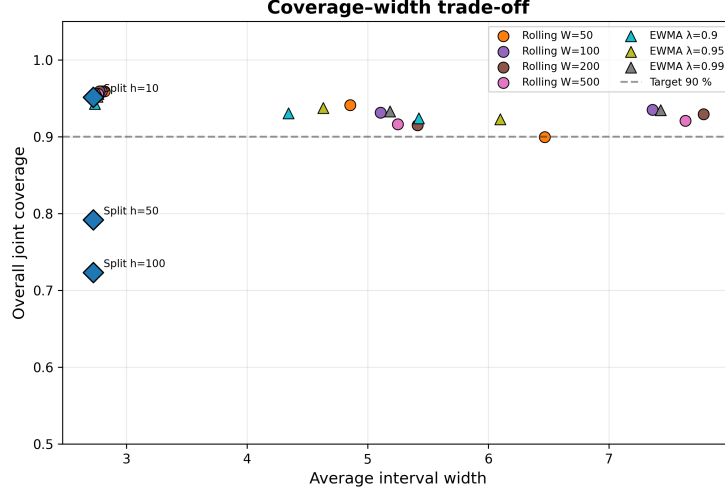


Figure 4: Coverage–width trade-off at horizon 100. Split conformal (blue diamonds) achieves narrow intervals but severe undercoverage. Adaptive methods (rolling window in circles, EWMA in triangles) form a Pareto frontier, trading width for reliable coverage. The gray dashed line marks the target 0.90 coverage level.

Coverage–Width Trade-off. This quantile expansion leads to a fundamental trade-off between coverage and interval width. Figure 4 illustrates this relationship at $h = 100$. Split conformal achieves narrow intervals but severe undercoverage, whereas adaptive methods form a Pareto frontier that trades increased width for reliable coverage.

For applications requiring strict coverage guarantees, rolling-window or EWMA calibration provides robust performance. More aggressive adaptation ($W = 50$, $\lambda = 0.90$) offers a favorable balance between coverage and efficiency, while conservative settings ($W \geq 200$, $\lambda \geq 0.95$) prioritize coverage stability at the cost of wider intervals.

Comparison with classical methods. To contextualize distribution-free uncertainty quantification, we compare with classical model-based approaches (Extended Kalman Filter) in Appendix C. We show that model-based methods suffer from severe undercoverage (joint coverage 0.736 vs 0.951 for split conformal at $h = 10$) due to violated Gaussian and linearization assumptions, highlighting the robustness of conformal methods under chaotic dynamics.

5 Discussion

5.1 Why Adaptive Conformal Works

Adaptive conformal methods succeed because they replace a static calibration distribution with a locally updated estimate of the error distribution, allowing prediction intervals to track the endogenous distribution shift induced by closed-loop forecasting. By continuously re-estimating quantiles from recent observations, adaptive methods maintain coverage guarantees even as the effective error distribution drifts, trading increased interval width for reliable uncertainty quantification under non-stationarity.

5.2 Practical Guidelines

Based on our empirical findings, we offer the following recommendations for practitioners:

- **Use adaptive conformal calibration** (rolling window or EWMA) when forecasts are deployed in closed-loop or when distribution shift is suspected. Static split conformal is unreliable beyond short horizons in chaotic or nonstationary systems.

- **Balance reactivity and stability** when selecting adaptive parameters. Smaller window sizes ($W \sim 50\text{--}100$) or decay factors ($\lambda \sim 0.90\text{--}0.95$) improve reactivity but increase variance; larger values provide smoother quantile evolution at the cost of slower adaptation.
- **Report joint coverage** for multivariate systems rather than per-dimension averages, as the latter can be misleadingly high even when joint coverage fails. Evaluate coverage across the full forecast horizon to detect degradation patterns.

5.3 Scope and Limitations

We note that if real-time model retraining or data assimilation were feasible, error accumulation in closed-loop forecasting could be mitigated. However, our study targets settings where forecasts are generated under a fixed model and cannot be revised retroactively—common in operational scenarios with computational constraints or when predictions must be committed in advance. In such settings, post-hoc uncertainty recalibration via adaptive conformal methods provides a principled approach to maintain coverage guarantees without modifying the forecasted trajectory.

For sufficiently long horizons, predictive uncertainty approaches the intrinsic scale of the Lorenz attractor. In this regime, maintaining nominal coverage necessarily requires intervals comparable to the attractor range—a consequence of the fundamental predictability limit of chaotic dynamics rather than a deficiency of adaptive conformal prediction. Accordingly, our empirical analysis focuses on horizons $h \in \{10, 50, 100\}$, where prediction intervals remain informative and the coverage–width trade-off is practically meaningful.

6 Conclusion

We investigated conformal prediction for uncertainty quantification in closed-loop forecasting of chaotic dynamical systems, using the Lorenz 63 system as a controlled testbed. Our results show that standard split conformal prediction suffers severe coverage degradation under closed-loop rollout due to endogenous distribution shift, despite strong performance in open-loop evaluation.

We demonstrated that adaptive conformal calibration methods, including rolling-window and EWMA schemes, substantially recover empirical coverage across long horizons by dynamically tracking the evolving error distribution. These gains come at the cost of wider prediction intervals, reflecting the intrinsic uncertainty of long-horizon chaotic forecasting. Overall, our findings highlight the necessity of adaptive uncertainty calibration when deploying learned predictors in closed-loop, nonstationary environments.

Broader Implications. While our study focuses on the Lorenz 63 system, the coverage degradation we observe is characteristic of closed-loop forecasting in any nonstationary or chaotic regime. Our findings extend to weather prediction, turbulent fluid dynamics, and autonomous control systems (settings where predictions must be committed without retroactive revision and where distribution shift is endemic). Adaptive conformal methods offer a principled, distribution-free approach to uncertainty calibration in closed-loop and nonstationary environments. Future work includes high-dimensional systems, regional conditional coverage, and integration with online learning.

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A System and Experimental Details

A.1 Lorenz 63 System Properties

The Lorenz 63 system [Lorenz, 1963] was originally derived as a simplified model of atmospheric convection. With parameters $\sigma = 10$, $\rho = 28$, and $\beta = 8/3$, the system exhibits:

- **Deterministic chaos:** Positive Lyapunov exponent $\lambda \approx 0.9$, meaning perturbations grow as $\sim e^{0.9t}$.
- **Strange attractor:** The dynamics evolve on a fractal set with dimension ≈ 2.06 .
- **Sensitivity to initial conditions:** Two trajectories starting 10^{-6} apart diverge to uncorrelated states within ~ 10 time units.

These properties make Lorenz 63 an ideal testbed: it is simple enough for extensive experimentation yet complex enough to exhibit the challenges of real-world chaotic forecasting.

A.2 Data Generation

We integrate the Lorenz equations using the fourth-order Runge-Kutta (RK4) method with time step $\Delta t = 0.01$. Initial conditions are sampled randomly on the attractor by: (1) integrating from arbitrary starting points for 1000 steps to reach the attractor, then (2) selecting $N_{\text{init}} = 10$ states separated by 100 time steps. We generate trajectories of $T = 10,000$ steps each and add i.i.d. Gaussian measurement noise $\epsilon_t \sim \mathcal{N}(0, \sigma_{\text{noise}}^2 I_3)$ with $\sigma_{\text{noise}} \in \{0.5, 1.0, 2.0\}$.

A.3 Adaptive Conformal Implementation Details

Update frequency. Both rolling and EWMA methods require recomputing the quantile at each time step, which can be computationally expensive for large W or long test sequences. In practice, we update the quantile every k steps (we use $k = 5$) rather than every step, using the same $\hat{q}_{1-\alpha}^{(t)}$ for multiple consecutive predictions. This reduces computation by 80% while maintaining reasonable tracking performance, as the error distribution typically does not shift drastically within 5 steps.

Parameter selection heuristics. For rolling window, window size W controls the reactivity–stability trade-off:

- $W = 50$: Highly reactive, tracks rapid distribution shifts but exhibits high variance in quantile estimates
- $W \in [100, 200]$: Balanced choice for moderate horizons ($h \sim 50$ – 100 steps)
- $W = 500$: Conservative, provides smooth quantile evolution but slower adaptation

For EWMA, decay parameter λ controls effective memory:

- $\lambda = 0.90$: Aggressive weighting of recent data, prioritizes narrow intervals
- $\lambda = 0.95$: Balanced default choice
- $\lambda = 0.99$: Near-uniform weighting, prioritizes coverage stability

The effective window size for EWMA is approximately $1/(1 - \lambda)$: $\lambda = 0.90$ corresponds to $W \approx 10$, while $\lambda = 0.99$ corresponds to $W \approx 100$.

B Additional Results and Analysis

B.1 Sensitivity to Noise Level

We study the sensitivity of predictive uncertainty to different levels of observation noise. Specifically, we consider Gaussian measurement noise with standard deviation $\sigma \in \{0.5, 1.0, 2.0\}$ while fixing the prediction horizon. For each noise level, we report the empirical coverage and average

interval width for EKF-based uncertainty estimates, split conformal prediction, and adaptive conformal methods.

As expected, higher noise levels lead to wider prediction intervals across all methods. While EKF intervals remain relatively narrow, they exhibit substantial under-coverage. In contrast, split conformal prediction maintains conservative coverage, and adaptive methods adjust interval widths dynamically to remain close to the nominal coverage level with reduced conservativeness compared to split conformal prediction.

B.2 Split Conformal Regional Performance

To assess spatial heterogeneity of uncertainty calibration, we analyze the regional performance of split conformal prediction over the Lorenz attractor. The (x, y) -plane is partitioned into a uniform 10×10 grid, and empirical coverage is computed separately within each region.

This regional analysis reveals pronounced variation in coverage across the attractor. In particular, regions corresponding to high-curvature or rapidly diverging trajectories exhibit more frequent coverage violations, while more stable regions tend to be over-covered. These effects are not visible in global coverage metrics and highlight the importance of regional diagnostics for chaotic dynamical systems.

B.3 Parameter Selection Guidelines

Based on the empirical results in Section 4, we provide practical guidance for selecting adaptive parameters:

Rolling window size W :

- $W = 50$: Highly reactive, best for rapidly changing environments. Achieves exactly the target coverage (0.900 at $h = 100$) with moderate width (6.47).
- $W = 100$: Balanced choice for most applications. Achieves highest coverage (0.935) with reasonable width (7.36).
- $W \geq 200$: Conservative, provides stable quantiles but slower adaptation. Diminishing returns beyond $W = 200$.

EWMA decay parameter λ :

- $\lambda = 0.90$: Aggressive weighting, prioritizes narrow intervals. Achieves 0.924 coverage with minimal width (5.42)—best efficiency.
- $\lambda = 0.95$: Balanced default choice (0.923 coverage, 6.10 width).
- $\lambda = 0.99$: Conservative weighting, maximizes coverage stability (0.934 coverage, 7.43 width).

Trade-off considerations: For safety-critical applications requiring strict coverage guarantees, use $W = 100$ or $\lambda \in [0.95, 0.99]$. For applications tolerating slight undercoverage in exchange for tighter intervals, use $\lambda = 0.90$ or $W = 50$. All adaptive configurations substantially outperform split conformal at long horizons.

C Comparison with Classical Uncertainty Methods

To contextualize the performance of conformal prediction, we compare split conformal intervals with the Extended Kalman Filter (EKF), a classical model-based approach to uncertainty quantification. The EKF provides a natural baseline for chaotic system forecasting: it explicitly models state uncertainty via a covariance matrix and propagates this uncertainty through the system dynamics via linearization.

C.1 Extended Kalman Filter Setup

The EKF estimates both the predicted state $\hat{\mathbf{x}}_{t+1}$ and an associated covariance matrix $\mathbf{P}_{t+1}$ representing uncertainty. At each time step, the EKF:

1. Predicts the next state using the learned LSTM dynamics: $\hat{\mathbf{x}}_{t+1} = f_{\text{LSTM}}(\mathbf{x}_t)$
2. Linearizes the dynamics via the Jacobian: $\mathbf{F}_t = \nabla f_{\text{LSTM}}(\mathbf{x}_t)$
3. Propagates uncertainty: $\mathbf{P}_{t+1} = \mathbf{F}_t \mathbf{P}_t \mathbf{F}_t^\top + \mathbf{Q}$
4. Constructs prediction intervals: $[\hat{x}_i \pm z_{1-\alpha/2} \sqrt{P_{ii}}]$ for each dimension i

The process noise $\mathbf{Q}$ and initial covariance $\mathbf{P}_0$ are tuned to match the empirical residual variance on the calibration set. We use the same $\alpha = 0.1$ significance level as conformal prediction for direct comparability.

C.2 Results and Discussion

Table 2 compares the EKF with split conformal prediction in both open-loop and closed-loop protocols at horizon $h = 10$. Figure 5 provides a visual comparison of coverage and interval width.

Table 2: Comparison of Extended Kalman Filter and split conformal prediction at horizon $h = 10$, $\sigma_{\text{noise}} = 0.5$. Joint coverage measures whether all three dimensions (x, y, z) simultaneously fall within their respective intervals. Per-dimension coverage reports the average coverage across dimensions independently.

Method	Joint Cov.	Per-dim Cov.	Width	RMSE	Assumptions
EKF	0.736	0.904	1.681	0.506	Gaussian, Linearization
Split (Open)	0.971	0.991	2.724	0.524	Distribution-free
Split (Closed)	0.951	0.983	2.724	0.553	Distribution-free

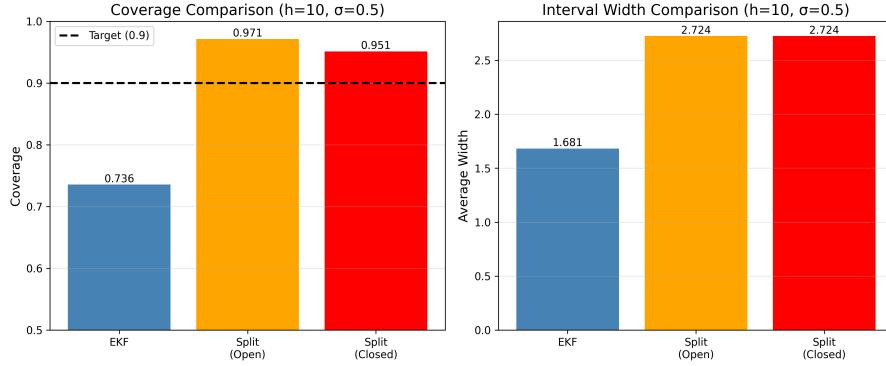


Figure 5: Comparison of Extended Kalman Filter (EKF) with split conformal prediction at horizon $h = 10$. Left: Joint coverage rates. The EKF achieves only 0.736 coverage despite Gaussian interval construction, falling significantly below the target 0.90. Split conformal achieves near-nominal coverage in both open-loop (0.971) and closed-loop (0.951) protocols. Right: Average interval widths. The EKF produces narrower intervals (1.681) than split conformal (2.724), but at the cost of severe undercoverage. All methods use the same significance level $\alpha = 0.1$.

Coverage Performance. The EKF achieves joint coverage of only 0.736, falling substantially below the target 0.90 level. While its per-dimension coverage of 0.904 appears acceptable, the joint coverage reveals that the three dimensions frequently fail to be covered simultaneously, indicating that the EKF’s Gaussian assumption fails to capture the true multivariate error distribution.

In contrast, split conformal achieves 0.971 coverage in open-loop and 0.951 in closed-loop at the same horizon, both comfortably exceeding the nominal target. This performance advantage stems from conformal prediction’s distribution-free nature: it makes no parametric assumptions about the error distribution and directly uses empirical residuals to construct intervals.

Interval Width. The EKF produces narrower intervals (average width 1.681) than split conformal (2.724), which might initially appear attractive. However, this narrow width comes at the cost of severe undercoverage: the EKF’s intervals fail to contain the true state nearly 30% of the time. The false precision of the EKF intervals reflects the breakdown of its Gaussian and linearity assumptions under chaotic dynamics—the true error distribution is heavy-tailed and skewed due to nonlinearity and chaos, which the EKF’s covariance propagation cannot capture.

Limitations of Model-Based Uncertainty. The EKF’s poor performance highlights fundamental limitations of model-based uncertainty quantification in chaotic systems. The EKF assumes: (1) Gaussian errors, enabling construction of intervals via $z_{1-\alpha/2}$ quantiles, which fails due to chaos-induced heavy tails; (2) accurate linearization via the Jacobian F_t , which breaks down as nonlinearities compound rapidly; and (3) correct uncertainty propagation via covariance updates, which relies on both Gaussian assumptions and linearization accuracy.

When these assumptions break down—as they inevitably do in chaotic forecasting—the EKF’s intervals become systematically miscalibrated. Conformal prediction sidesteps these issues entirely by constructing intervals directly from empirical residuals, making no assumptions about functional form or dynamics. While the EKF could in principle be made adaptive by updating Q online, this would still inherit the Gaussian and linearization assumptions. In contrast, adaptive conformal prediction (Section 6) adapts to distribution shift without parametric constraints, providing a more robust foundation for uncertainty quantification under non-stationarity.

D Implementation Details

D.1 LSTM Training Details

Model architecture. The forecasting model is a two-layer LSTM with hidden size 128 and output dimension 3, corresponding to the Lorenz system state variables (x, y, z) . Inputs consist of sliding windows of length $k = 50$ consecutive observations, and the model is trained to predict the next state one step ahead: $\hat{x}_{t+1} = f_{\text{LSTM}}(\{x_{t-k+1}, \dots, x_t\})$.

Training procedure. The model is trained using a one-step-ahead prediction objective with *teacher forcing*, where true observations from the training set are provided as inputs at each step. This supervised learning approach optimizes the model’s performance on single-step predictions given accurate state information. Training is performed using the Adam optimizer with a fixed learning rate of 10^{-3} , batch size of 64, and early stopping based on validation loss to prevent overfitting. All input data are standardized using statistics (mean and standard deviation) computed from the training set only; the same transformation is applied to calibration and test data. Random seeds are fixed throughout to ensure reproducibility.

Rollout strategies. During evaluation, we distinguish between two rollout strategies that differ in how they consume model predictions:

- **Open-loop rollout:** Each prediction step uses the true observed state from the test set as input, maintaining consistency with the training distribution. This represents an idealized scenario where the model has access to ground truth at every step.
- **Closed-loop rollout:** The model’s own predictions are recursively fed back as inputs for subsequent predictions, without access to intermediate ground truth observations. Starting from an initial true state x_t , the model generates $\hat{x}_{t+1}, \hat{x}_{t+2}, \dots, \hat{x}_{t+h}$ by iteratively applying $\hat{x}_{t+i} = f_{\text{LSTM}}(\{\hat{x}_{t+i-k}, \dots, \hat{x}_{t+i-1}\})$.

Distribution mismatch. Critically, closed-loop forecasting is applied *only at evaluation time*, creating a distribution mismatch between training and deployment. The model is optimized to perform well when conditioned on true states drawn from the data distribution, but during closed-loop evaluation it must operate on its own potentially erroneous predictions. This discrepancy between the training distribution (true states) and the closed-loop evaluation distribution (model-generated states) plays a central role in the breakdown of uncertainty calibration: as prediction errors compound over the forecast horizon, the effective error distribution shifts away from what was observed during calibration, violating the exchangeability assumption required by standard conformal prediction.

D.2 Code Availability

All experiments in this paper are fully reproducible. The complete codebase, including data generation, model training, rollout evaluation, and conformal calibration, is publicly available at:

<https://github.com/DerekTsai7113/adaptive-conformal-lorenz63>